

TRANSFER OF KOSZULNESS

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RTG Seminar April 13, 2026

DISCLAIMER

These are the notes for my talk in the RTG Seminar on 4/13/26. These notes mostly follow section 3.2 of the survey paper “Koszul algebras and Regularity” by Conca, De Negri, and Rossi, henceforth abbreviated to [C,DN,R]. Though we occasionally fill in some details, **nothing here is my original work**. Thanks are due to Adam LaClair and Mark Walker for helping me work through many of these details.

Standing Assumptions: Throughout, k is a field, k -algebras A are standard $\mathbb{N}$ -graded with $A_0 = k$, and A_1 a finite-dimensional k -vector space. Denote the homogeneous maximal ideal of A by $\mathfrak{m}_A$, or $\mathfrak{m}$ when there’s no risk of confusion. Modules and ideals are assumed graded and finitely generated, and all module homomorphisms have degree zero.

Transfer Between Quotients

Question 1. *Consider a k -algebra A and a quotient algebra $B = A/I$. If one of A or B is Koszul, when is the other also Koszul?*

Pop Quiz: Since we want $B = A/I$ to inherit the standard grading from A , what are we implicitly assuming about the ideal I ?

As one might be able to surmise from the title of the relevant paper, sufficient conditions to an answer for Question 1 lie in appropriate bounds on the (Castelnuovo-Mumford) regularities.

Recall. For an A -module M we put

$$t_i^A(M) = \sup\{j : \beta_{ij}^A(M) \neq 0\} \text{ and } \operatorname{reg}_A M = \sup\{t_i^A(M) - i : i \geq 0\}.$$

Before we give an answer to Question 1, we need a couple of results. The paper proves a version of the following lemma for arbitrary complexes, but for our purposes we can get away with the exact version.

Lemma 1 (2.9(2) in [C,DN,R]). *Let $M_\bullet = \cdots \rightarrow M_2 \rightarrow M_1 \rightarrow M_0 \rightarrow N \rightarrow 0$ be an exact complex of A -modules. Then*

$$t_i^A(N) \leq \max\{t_j^A(M_{i-j}) : j = 0, \dots, i\}.$$

In particular, we have

$$\operatorname{reg}_A(N) \leq \sup\{\operatorname{reg}_A(M_j) - j : j \geq 0\}.$$

Proof. We establish the first inequality by inducing on i . For $i = 0$, the max on the RHS is irrelevant. Since $- \otimes_A k$ is right exact, exactness of $M_0 \rightarrow N \rightarrow 0$ gives us an exact sequence $(M_0 \otimes_A k)_j \rightarrow (N \otimes_A k)_j \rightarrow 0$ for each graded piece j [because our maps preserve degree by assumption, and thus so do the induced maps on the Tor’s]. So whenever the j th graded piece of $N \otimes_A k$ is nonzero, so is the j th graded piece of $M_0 \otimes_A k$. It follows that $t_0^A(N) \leq t_0^A(M_0)$.

For $i > 0$, we put $K = \ker(M_0 \rightarrow N)$ giving us the short exact sequence

$$0 \rightarrow K \rightarrow M_0 \rightarrow N \rightarrow 0.$$

Notice that since $K = \operatorname{im}(M_1 \rightarrow M_0)$, we have the exact sequence

$$\cdots \rightarrow M_2 \rightarrow M_1 \rightarrow K \rightarrow 0.$$

The long exact sequence in Tor gives us

$$\mathrm{Tor}_i(M_0, k)_j \rightarrow \mathrm{Tor}_i(N, k)_j \rightarrow \mathrm{Tor}_{i-1}(K, k)_j$$

for each j th graded piece. Whenever the middle term is nonzero, at least one of the other ones must also be nonzero. Hence, we obtain

$$\begin{aligned} t_i^A(N) &\leq \max\{t_i^A(M_0), t_{i-1}^A(K)\} \\ &\leq \max\{t_i^A(M_0), \max\{t_j^A(M_{(i-1)-j+1}) : j = 0, \dots, i-1\}\} \\ &= \max\{t_j^A(M_{i-j}) : j = 0, \dots, i\}. \end{aligned} \tag{IH}$$

A word on the indices when we invoke the inductive hypothesis: For the inductive step, we replace i by $i-1$ in the above statement, but now we're treating K as N in the exact complex, hence the indices start at 1 and not 0, so we add another 1 back. Hence, the first inequality follows from induction.

Finally, we deduce that

$$\begin{aligned} \mathrm{reg}_A(N) &= \sup_i \{t_i^A(N) - i\} \\ &\leq \sup_i \{\max\{t_j^A(M_{i-j}) : j = 0, \dots, i\} - i\} \\ &= \sup_i \{\max\{t_{i-\ell}^A(M_\ell) : \ell = 0, \dots, i\} - (i - \ell) - \ell\} \quad (\text{Put } \ell = i - j.) \\ &\leq \sup_\ell \{\sup_i \{t_{i-\ell}^A(M_\ell) - (i - \ell)\} - \ell\} \\ &= \sup_\ell \{\mathrm{reg}_A(M_\ell) - \ell\}. \end{aligned}$$

□

With this lemma, we can prove the following technical result that will do a lot of the heavy lifting for us in a bit.

Proposition (3.3 in C,DN,R). *Let A be a k -algebra and B a quotient algebra of A . Let M be a B -module. Then:*

- (i) $\mathrm{reg}_A(B) \leq 1 \implies \mathrm{reg}_B(M) \leq \mathrm{reg}_A(M)$, and
- (ii) $\mathrm{reg}_A(M) \leq \mathrm{reg}_A(B) + \mathrm{reg}_B(M)$.

Proof. (i) Assume $\mathrm{reg}_A(B) \leq 1$. Notice it is enough to show that for all i , one has

$$t_i^B(M) - i \leq \max\{t_j^A(M) - j : j = 0, 1, \dots, i\}, \tag{†}$$

because then taking the sup on both sides over all i gives us the desired inequality. We argue by induction on i . When $i = 0$, the max on the RHS is irrelevant and $t_0^A(M)$ and $t_0^B(M)$ both count the largest degree of a minimal generator of M , giving us the equality $t_0^A(M) = t_0^B(M)$. Now take $i > 0$, assume the result holds for $i-1$. Then take a minimal presentation for M as a B -module

$$0 \rightarrow K \rightarrow F \rightarrow M \rightarrow 0.$$

Tensoring over B against k , the relevant LES in Tor is

$$\mathrm{Tor}_i^B(F, k)_j \rightarrow \mathrm{Tor}_i^B(M, k)_j \rightarrow \mathrm{Tor}_{i-1}^B(K, k)_j.$$

Since F is free over B , for $i > 0$ the leftmost Tor module vanishes and so the remaining (possibly) nonzero map is an injection. So whenever the middle Tor is not zero, the right most Tor must also be nonzero. Hence, we have the estimate $t_i(M) \leq t_{i-1}(K)$ for each $i > 0$. Therefore, one can invoke the inductive hypothesis to see that

$$t_i^B(M) - i \leq t_{i-1}^B(K) - i = t_{i-1}(K) - (i-1) - 1 \stackrel{IH}{\leq} \max\{t_j^A(K) - j - 1 : j = 0, \dots, i-1\}.$$

When we consider $0 \rightarrow K \rightarrow F \rightarrow M \rightarrow 0$ as an SES of A -modules via restriction of scalars, the induced LES in Tor

$$\mathrm{Tor}_{j+1}^A(M, k) \rightarrow \mathrm{Tor}_j^A(K, k) \rightarrow \mathrm{Tor}_j^A(F, k)$$

gives us $t_j^A(K) \leq \max\{t_j^A(F), t_{j+1}^A(M)\}$. Moreover, we see that $F = \bigoplus_{\ell} B(-\ell)^{\beta_{0\ell}^B(M)}$ and so

$$t_j^A(F) = t_j^A\left(\bigoplus_{\ell} B(-\ell)^{\beta_{0\ell}^B(M)}\right) = t_j^A(B) + t_0^A(M) \underset{\mathrm{reg}_A B \leq 1}{\leq} j + 1 + t_0^A(M).$$

With these estimates, we find that

$$\begin{aligned} t_i^B(M) - i &\leq \max\{t_j^A(K) - j - 1 : j = 0, \dots, i - 1\} \\ &\leq \max\{\max\{j + 1 + t_0^A(M), t_{j+1}^A(M)\} - j - 1 : j = 0, \dots, i - 1\} \\ &= \max\{t_0^A(M), \max\{t_{j+1}^A(M) - j - 1 : j = 0, \dots, i - 1\}\} \\ &\leq \max\{t_j^A(M) - j : j = 0, \dots, i\}. \end{aligned}$$

Thus, $(\dagger)$ holds by induction, and taking sups over all i on both sides gives the desired inequality regarding regularity.

(ii) Consider $F_{\bullet} = \dots \rightarrow F_1 \rightarrow F_0 \rightarrow M \rightarrow 0$, the (augmented) minimal free resolution of M as an B -module. Then $F_{\bullet}$ is an exact complex of A -modules [via restriction of scalars along $A \rightarrow A/I = B$], whence Lemma 1 implies that

$$\mathrm{reg}_A(M) \leq \sup\{\mathrm{reg}_A(F_j) - j : j \geq 0\}.$$

Notice now that $\mathrm{reg}_A B(-\ell) = \mathrm{reg}_A B + \ell$. And since $F_{\bullet}$ is the minimal B -free resolution of M , we have $F_j = \bigoplus_{\ell} B(-\ell)^{\beta_{j\ell}^B(M)}$, whence

$$\begin{aligned} \mathrm{reg}_A(F_j) &= \mathrm{reg}_A\left(\bigoplus_{\ell} B(-\ell)^{\beta_{j\ell}^B(M)}\right) \\ &= \mathrm{reg}_A(B) + \sup\{\ell \mid \beta_{j\ell}^B(M) \neq 0\} \\ &= \mathrm{reg}_A(B) + t_j^B(M). \end{aligned}$$

Therefore, from our previous inequality obtained from Lemma 1, we deduce

$$\begin{aligned} \mathrm{reg}_A(M) &\leq \sup\{\mathrm{reg}_A(B) + t_j^B(M) - j : j \geq 0\} \\ &= \mathrm{reg}_A(B) + \sup\{t_j^B(M) - j : j \geq 0\} \\ &= \mathrm{reg}_A(B) + \mathrm{reg}_B(M). \end{aligned}$$

□

Throughout this seminar, Ryan and Kesavan proved (parts of) the following result of Avramov-Eisenbud-Peeva [analogous to the classical Auslander-Buchsbaum-Serre RLR characterizations].

Theorem (2.7 in C, DN, R). *[Avramov-Eisenbud-Peeva] TFAE:*

1. A is Koszul,
2. $\mathrm{reg}_A(k) < \infty$,
3. $\mathrm{reg}_A(M) < \infty$ for every A -module M .

Proof. Exercise. □

Now an answer to our question follows rather easily from the Proposition and Lemma 1.

Theorem 1 (3.2 in C, DN, R). *Let A be a k -algebra and $B = A/I$ a quotient algebra. Then,*

(1) *If A is Koszul and $\text{reg}_A(B) \leq 1$, then B is Koszul.*

(2) *If B is Koszul and $\text{reg}_A(B) < \infty$, then A is Koszul.*

Proof. (1). Suppose A is Koszul and $\text{reg}_A(B) \leq 1$. Then Prop (i) with $M = k$ gives us $\text{reg}_B(k) \leq \text{reg}_A(k)$, and $\text{reg}_A(k) = 0$ since A is Koszul. We conclude $\text{reg}_B(k) = 0$ and thus B is Koszul.

(2). Assume B is Koszul and $\text{reg}_A(B) < \infty$. Putting $M = k$ once again, Lemma 1(ii) shows

$$\text{reg}_A(k) \leq \underbrace{\text{reg}_B(k)}_{< \infty \text{ b/c } B \text{ Koszul} + \text{THRM A-E-P}} + \underbrace{\text{reg}_A(B)}_{< \infty \text{ by assumption}} < \infty$$

In fact, $\text{reg}_B(k) = 0 : B$ is Koszul means $\beta_{ij}^B(k) = 0$ iff $i \neq j$, so $\text{reg}_B(k) = \sup\{t_i^B(k) - i \mid i \in \mathbb{N}\} = \sup\{j - i \mid \beta_{ij}^B(k) \neq 0\}$. Thus, A is Koszul by Theorem [A-E-P]. $\square$

Transfer to Veronese Submodules

Definition. Recall that for $c \in \mathbb{N}$, we write $A^{(c)}$ for the c -th Veronese subring of A . More generally, for a graded A -module [recall our standing assumptions] we put define the c -th Veronese submodule of M as

$$M^{(c)} := \bigoplus_{j \in \mathbb{Z}} M_{jc}.$$

We also put

$$V_u = \bigoplus_{j \in \mathbb{Z}} A_{jc+u}$$

for an integer u with $0 \leq u \leq c-1$. Notice V_u is an $A^{(c)}$ -module generated [as an $A^{(c)}$ -module] in degree zero [with the grading given by the above direct sum decomposition].

Remark.

1. We have an equality

$$A(-j)^{(c)} = V_u(-[j/c]) \tag{\dagger}$$

where $u = 0$ if $j \equiv 0 \pmod{c}$ and $u = c - r$ if $j \equiv r \pmod{c}$ and $0 < r < c$.

2. Since our modules are standard graded f.g. and the R -module homomorphisms have degree 0, the formation of the (c) -th Veronese is an exact functor

$$(-)^{(c)} : \mathbf{A} - \mathbf{Mod} \rightarrow \mathbf{A}^{(c)} - \mathbf{Mod}.$$

One can see this by just considering the graded pieces of each module in a short exact sequence and using the fact that the morphisms are degree preserving. You're exact if and only if your restriction to each graded piece is also exact.

Fact (3.5 in C,DN,R). *If A is Koszul, then $\text{reg}_{A^{(c)}}(V_u) = 0$ for all $u = 0, \dots, u-1$.*

The proof requires another lemma we won't have time to prove, and most of it is captured in spirit in the proof of the next result. As such, we shall skip the proof of this fact. Now one might wonder:

Question 2. *If A is Koszul, are the Veronese subalgebras $A^{(c)}$ also Koszul?*

Theorem 2 (Also 3.5 in C,DN,R). *If A is a Koszul, then $A^{(c)}$ is also Koszul.*

Proof. Consider the augmented minimal free resolution of k over A , call it $F_\bullet$. Applying the exact functor $(-)^{(c)}$, we have $H_i(F_\bullet^{(c)}) = 0$ for all i as well. Since A is Koszul, $\beta_{ij}(k) = 0$ whenever $i \neq j$, so from the equality $F_j = \bigoplus^{\beta_{jj}(k)} A(-j)$ we see that

$$F_j^{(c)} = \bigoplus^{\beta_{jj}(k)} A(-j)^{(c)} = \bigoplus^{\beta_{jj}(k)} V_u(-\lceil j/c \rceil)$$

with $u = 0$ if $j \equiv 0 \pmod{c}$ and $u = c - r$ if $j \equiv r \pmod{c}$ with $0 < r < c$. In any case, by the Fact, we see that $\text{reg}_{A^{(c)}} V_u(-\lceil j/c \rceil) = \lceil j/c \rceil$. Now by Lemma 1 we obtain

$$\text{reg}_{A^{(c)}} k \leq \sup\{\lceil j/c \rceil - j : j \geq 0\} = 0$$

because $\lceil j/c \rceil \leq j \implies \lceil j/c \rceil - j \leq 0$ for each $j \in \mathbb{N}$, and when $j = 0$, we actually obtain the value zero. Thus, $A^{(c)}$ is Koszul. $\square$

Theorem 3 (3.6 in C,DN,R). *For any k -algebra R , $R^{(c)}$ is Koszul for $c \gg 0$. More specifically, if $R = A/I$ with A Koszul, then $R^{(c)}$ is Koszul for every $c \geq s := \sup\{t_i^A(R)/(1+i) : i \geq 0\}$.*

Proof. Let $R = A/I$ with A Koszul [and I homogeneous of course]. Let $F_\bullet$ be the augmented minimal free resolution of R as an A -module. Note that $A^{(c)}$ is Koszul [for any $c \geq 1$] by the Theorem 2. Then since $(-)^{(c)}$ is an exact functor, the complex $F_\bullet^{(c)}$ is an exact complex of $A^{(c)}$ -modules.

Moreover, for $i \geq 0$, $F_i^{(c)}$ is a direct sum of shifted copies of the Veronese submodules V_u as we see from $(\ddagger)$. Since $\text{reg}_{R^{(c)}}(V_u) = 0$ by the Fact, and the F_i are shifted by at most $t_i^A(R)$, we obtain the bound on the regularity of $F_i^{(c)}$

$$\text{reg}_{A^{(c)}} F_i^{(c)} \leq \lceil t_i^A(R)/c \rceil \quad (\star)$$

in light of $(\ddagger)$ [and the fact that the $F_i^{(c)}$ is free over $R^{(c)}$, so its regularity truly is determined by the maximal shift of the copies of V_u]. Now noticing $F_{-1}^{(c)} = R^{(c)}$, invoking the Lemma 1 gives us

$$\begin{aligned} \text{reg}_{A^{(c)}} R^{(c)} &\leq \sup\{\text{reg}_{A^{(c)}}(F_i^{(c)}) - i : i \geq 0\} \\ &\leq \sup\{\lceil t_i^A(R)/c \rceil - i : i \geq 0\}. \end{aligned} \quad (\text{by } (\star))$$

Now notice that whenever $c \geq t_i^A(R)/(1+i)$, we have that

$$\frac{t_i^A(R)}{c} \leq i+1 \implies \left\lceil \frac{t_i^A(R)}{c} \right\rceil \leq i+1.$$

Whence, putting $c \geq s = \sup\{t_i^A(R)/(1+i) : i \geq 0\}$, our above bound [that followed from $(\star)$] gives us

$$\text{reg}_{A^{(c)}}(R^{(c)}) \leq 1 \xrightarrow{\text{Thm. 1}} R^{(c)} \text{ is Koszul.}$$

$\square$

I expect this is where we will end due to time.

If time permits, we prove the following lemma, in hopes of proving the Fact afterward.

Lemma 2 (3.4 in C,DN,R). *Suppose R is a Koszul algebra and M is an R -module. Then*

$$\operatorname{reg}_R(\mathfrak{m}M) \leq \operatorname{reg}_R(M) + 1.$$

In particular, $\operatorname{reg}_R(\mathfrak{m}^u) = u$ unless $\mathfrak{m}^u = 0$. That is, $\mathfrak{m}^u$ has a linear resolution for every $u \in \mathbb{N}$.

Proof. The first assertion follows from the usual argument we've been giving throughout: The SES

$$0 \rightarrow \mathfrak{m}M \rightarrow M \rightarrow M/\mathfrak{m}M \rightarrow 0$$

induces an LES in Tor

$$\cdots \rightarrow \operatorname{Tor}_{i+1}(M/\mathfrak{m}M, k)_j \rightarrow \operatorname{Tor}_i(\mathfrak{m}M, k)_j \rightarrow \operatorname{Tor}_i(M, k)_j \rightarrow \cdots$$

whence $\operatorname{Tor}_i(\mathfrak{m}M, k)_j \neq 0$ implies that one of the other Tor's above is also nonzero, so

$$t_i^R(\mathfrak{m}M) \leq \max\{t_{i-1}^R(M/\mathfrak{m}M) + 1, t_i^R(M)\},$$

and taking the sup over all i yields

$$\operatorname{reg}_R(\mathfrak{m}M) \leq \max\{\operatorname{reg}_R(M/\mathfrak{m}M) + 1, \operatorname{reg}_R(M)\}. \quad (\star\star)$$

Now notice that since $- \otimes_R k$ is right exact, the end of the minimal free resolution of M over R leads us to an onto map of k -vector spaces as seen below.

$$\bigoplus_j R(-j)^{\beta_{0j}(M)} \twoheadrightarrow M \rightarrow 0 \xrightarrow{- \otimes_R k} \bigoplus_j k(-j)^{\beta_{0j}(M)} \twoheadrightarrow M/\mathfrak{m}M \rightarrow 0.$$

As M is finitely generated, $\beta_{0j}(M) < \infty$ for each j , so the last onto map is an onto map of finite dimensional k -vector spaces, and thus an isomorphism. The upshot is that $M/\mathfrak{m}M$ is a direct sum of copies of k shifted by at most $-t_0(M)$. This along with $(\star\star)$ gives us the desired inequality

$$\operatorname{reg}_R(\mathfrak{m}M) \leq \operatorname{reg}_R(M) + 1.$$

Now for the second assertion, we induce on u . When $u = 1$, we see that $\mathfrak{m} = 0 \oplus R_1 \oplus R_2 \oplus \cdots$ is minimally generated as an R -module in degree one, and thus $\operatorname{reg}_R(\mathfrak{m}) \geq 1$. On the other hand, putting $M = R$ in what we just proved, we have $\operatorname{reg}_R(\mathfrak{m}) \leq \operatorname{reg}_R(R) + 1 = 1$ since R is minimally generated by $1 \in R_0$ as a module over itself [so $\operatorname{reg}_R R = 0$].

Now for $u > 1$ we put $M = \mathfrak{m}^{u-1}$ in what we just proved to deduce that $\operatorname{reg}(\mathfrak{m}^u) \leq \operatorname{reg}(\mathfrak{m}^{u-1}) + 1 = u$ by the (unstated) inductive hypothesis. On the other hand, $\mathfrak{m}^u = \bigoplus_0 \oplus_1 \oplus \cdots \oplus \bigoplus_{u-1} \oplus R_u \oplus R_{u+1} \oplus \cdots$ is minimally generated as an R -module in degree u , so in fact $\operatorname{reg}_R(\mathfrak{m}^u) = u$. $\square$

Now we are equipped to prove the Fact.

Proof Sketch of Fact 2. Show by induction on i that $t_i^{A^{(c)}}(V_u) \leq i$ for every i . The $i = 0$ case is obvious. For $i > 0$ apply Lemma 2 with $M = \mathfrak{m}^u(u)$, and notice $\operatorname{reg} M = 0$ by Lemma 2 as well. Take the augmented minimal free resolution $F_\bullet$ of M as an A -module, apply the c th Veronese functor to get an exact complex $F_\bullet^{(c)}$ of $A^{(c)}$ -modules. Notice $F_j^{(c)}$ is a direct sum of $R(-j)^{(c)}$'s and invoke Lemma 1 to get an upper bound on $t_i^{A^{(c)}}$. Notice $F_0^{(c)}$ is free over $A^{(c)}$, so $\operatorname{reg}_{A^{(c)}} F_0 = -\infty$. For $j > 0$, we see $A(-j)^{(c)} = V_w(-\lceil j/c \rceil)$ so that by induction we get $t_{i-j}^{A^{(c)}}(F_j^{(c)}) \leq i - j + \lceil j/c \rceil \leq i$. From this, one can obtain

$$t_i^{A^{(c)}}(V_u) \leq \max\{i - j + \lceil j/c \rceil : j = 1, \dots, i\} = i.$$

$\square$